

Homework 5

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Problem 1

Given: $N_A = 10^{14} \text{ cm}^{-3}$ doped p-type Si bar (semi-infinite, $x \geq 0$), dark, room temperature ($T = 300 \text{ K}$), $n_i = 10^{10} \text{ cm}^{-3}$. A minority carrier digester at $x = 0$ makes $n = 0$ at $x = 0$.

(a) Equilibrium conditions (no digester)

Since the bar is p-type with $N_A = 10^{14} \text{ cm}^{-3} \gg n_i = 10^{10} \text{ cm}^{-3}$:

(i) Majority carrier concentration:

$$p_0 = N_A = 10^{14} \text{ cm}^{-3}$$

(ii) Minority carrier concentration:

$$n_0 = \frac{n_i^2}{p_0} = \frac{(10^{10})^2}{10^{14}} = \frac{10^{20}}{10^{14}} = 10^6 \text{ cm}^{-3}$$

(b) δn_p at $x = 0$ with digester on

The digester forces $n = 0$ at $x = 0$. The total electron concentration is $n = n_0 + \delta n_p$. Therefore:

$$0 = n_0 + \delta n_p(0) \implies \delta n_p(0) = -n_0$$

$$\delta n_p(0) = -10^6 \text{ cm}^{-3}$$

(c) Low-level injection?

Low-level injection requires $|\delta n_p| \ll p_0$:

$$|\delta n_p(0)| = 10^6 \text{ cm}^{-3} \ll p_0 = 10^{14} \text{ cm}^{-3}$$

Since $10^6 \ll 10^{14}$ (eight orders of magnitude smaller):

Yes, low-level injection conditions are satisfied.

(d) $\delta n_p(x)$ under steady state

Given: $\mu_n = 1000 \text{ cm}^2/\text{Vs}$, $\tau_n = 1 \text{ } \mu\text{s} = 10^{-6} \text{ s}$.

First, compute the diffusion coefficient and diffusion length:

$$D_n = \frac{kT}{q} \mu_n = (0.02585)(1000) = 25.85 \text{ cm}^2/\text{s}$$

$$L_n = \sqrt{D_n \tau_n} = \sqrt{25.85 \times 10^{-6}} = \sqrt{2.585 \times 10^{-5}} = 5.084 \times 10^{-3} \text{ cm} \approx 50.8 \text{ } \mu\text{m}$$

Under steady state with no generation ($G_L = 0$, dark) and no electric field (uniform doping), the minority carrier diffusion equation is:

$$D_n \frac{d^2(\delta n_p)}{dx^2} - \frac{\delta n_p}{\tau_n} = 0$$

The general solution is:

$$\delta n_p(x) = A e^{-x/L_n} + B e^{+x/L_n}$$

Boundary conditions:

- As $x \rightarrow \infty$: $\delta n_p \rightarrow 0$, so $B = 0$.
- At $x = 0$: $\delta n_p(0) = -10^6 \text{ cm}^{-3}$, so $A = -10^6 \text{ cm}^{-3}$.

$$\delta n_p(x) = -10^6 e^{-x/L_n} \text{ cm}^{-3}, \quad L_n = 5.084 \times 10^{-3} \text{ cm} \approx 50.8 \text{ } \mu\text{m}$$

(e) Minority carrier current at $x = 0$

In the absence of an electric field, the minority carrier current is purely diffusion:

$$J_n = q D_n \frac{d(\delta n_p)}{dx}$$

Computing the derivative:

$$\frac{d(\delta n_p)}{dx} = -10^6 \left(-\frac{1}{L_n} \right) e^{-x/L_n} = \frac{10^6}{L_n} e^{-x/L_n}$$

At $x = 0$:

$$\left. \frac{d(\delta n_p)}{dx} \right|_{x=0} = \frac{10^6}{5.084 \times 10^{-3}} = 1.967 \times 10^8 \text{ cm}^{-4}$$

$$J_n(0) = q D_n \times 1.967 \times 10^8 = (1.6 \times 10^{-19})(25.85)(1.967 \times 10^8)$$

$$J_n(0) = 8.14 \times 10^{-10} \text{ A/cm}^2 \quad (\text{in the } +x \text{ direction})$$

The current is in the $+x$ direction because electrons diffuse toward lower concentration (toward $x = 0$), so electron particle flow is in the $-x$ direction, and conventional current is in the $+x$ direction.

Problem 2

Show that $\nabla \cdot \mathbf{J} = \nabla \cdot (\mathbf{J}_n + \mathbf{J}_p) = 0$ under steady-state conditions (including illumination).

The continuity equations for electrons and holes are:

$$\frac{\partial n}{\partial t} = \frac{1}{q} \nabla \cdot \mathbf{J}_n + G_n - R_n$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \nabla \cdot \mathbf{J}_p + G_p - R_p$$

Under **steady-state** conditions, $\frac{\partial n}{\partial t} = 0$ and $\frac{\partial p}{\partial t} = 0$. Therefore:

$$\frac{1}{q} \nabla \cdot \mathbf{J}_n = R_n - G_n \tag{1}$$

$$-\frac{1}{q} \nabla \cdot \mathbf{J}_p = R_p - G_p \implies \frac{1}{q} \nabla \cdot \mathbf{J}_p = G_p - R_p \tag{2}$$

Adding equations (1) and (2):

$$\frac{1}{q} \nabla \cdot (\mathbf{J}_n + \mathbf{J}_p) = (R_n - G_n) + (G_p - R_p)$$

Since electrons and holes are always generated and recombine in **pairs** (each generation event creates one electron and one hole, each recombination event destroys one of each):

$$G_n = G_p \equiv G \quad \text{and} \quad R_n = R_p \equiv R$$

Substituting:

$$\frac{1}{q} \nabla \cdot (\mathbf{J}_n + \mathbf{J}_p) = (R - G) + (G - R) = 0$$

Therefore:

$$\boxed{\nabla \cdot \mathbf{J} = \nabla \cdot (\mathbf{J}_n + \mathbf{J}_p) = 0} \quad \square$$

This result holds regardless of the generation rate (including optical illumination), as long as the system is in steady state.

Problem 3

Given: Si, $N_D = 10^{15} \text{ cm}^{-3}$, $G_L = 10^{19} \text{ cm}^{-3}/\text{s}$ (uniform), $\tau_n = \tau_p = 10 \text{ } \mu\text{s} = 10^{-5} \text{ s}$, $D_p = 12 \text{ cm}^2/\text{s}$, $T = 300 \text{ K}$.

Equilibrium values:

$$n_0 = N_D = 10^{15} \text{ cm}^{-3}, \quad p_0 = \frac{n_i^2}{N_D} = \frac{10^{20}}{10^{15}} = 10^5 \text{ cm}^{-3}$$

Excess carrier concentrations (uniform illumination $\Rightarrow$ no spatial gradients):

$$\delta n = \delta p = G_L \cdot \tau = 10^{19} \times 10^{-5} = 10^{14} \text{ cm}^{-3}$$

Total carrier concentrations:

$$n = n_0 + \delta n = 10^{15} + 10^{14} = 1.1 \times 10^{15} \text{ cm}^{-3}$$

$$p = p_0 + \delta p = 10^5 + 10^{14} \approx 10^{14} \text{ cm}^{-3}$$

(a) Quasi-Fermi level separation $F_n - F_p$

Using the quasi-Fermi level relations:

$$n = n_i \exp\left(\frac{F_n - E_i}{kT}\right), \quad p = n_i \exp\left(\frac{E_i - F_p}{kT}\right)$$

Multiplying:

$$np = n_i^2 \exp\left(\frac{F_n - F_p}{kT}\right)$$

Solving for the quasi-Fermi level separation:

$$\begin{aligned} F_n - F_p &= kT \ln\left(\frac{np}{n_i^2}\right) \\ np &= (1.1 \times 10^{15})(10^{14}) = 1.1 \times 10^{29} \text{ cm}^{-6} \\ F_n - F_p &= (0.02585) \ln\left(\frac{1.1 \times 10^{29}}{10^{20}}\right) = (0.02585) \ln(1.1 \times 10^9) \\ &= (0.02585)(0.0953 + 9 \times 2.3026) = (0.02585)(20.819) \\ &\quad \boxed{F_n - F_p \approx 0.538 \text{ eV}} \end{aligned}$$

(b) Change in conductivity

The mobility for holes from Einstein relation:

$$\mu_p = \frac{qD_p}{kT} = \frac{D_p}{kT/q} = \frac{12}{0.02585} = 464 \text{ cm}^2/\text{Vs}$$

Using a standard Si electron mobility of $\mu_n \approx 1350 \text{ cm}^2/\text{Vs}$ (for $N_D = 10^{15} \text{ cm}^{-3}$ at 300 K):

$$\Delta\sigma = q(\delta n \mu_n + \delta p \mu_p) = q \cdot \delta n(\mu_n + \mu_p)$$

since $\delta n = \delta p = 10^{14} \text{ cm}^{-3}$:

$$\begin{aligned} \Delta\sigma &= (1.6 \times 10^{-19})(10^{14})(1350 + 464) \\ &= (1.6 \times 10^{-5})(1814) \end{aligned}$$

$$\boxed{\Delta\sigma \approx 0.0290 (\Omega \cdot \text{cm})^{-1}}$$

(c) Band diagram with quasi-Fermi levels

Compute the positions of F_n and F_p relative to E_i :

$$F_n - E_i = kT \ln\left(\frac{n}{n_i}\right) = 0.02585 \ln\left(\frac{1.1 \times 10^{15}}{10^{10}}\right) = 0.02585 \ln(1.1 \times 10^5) = 0.02585 \times 11.608 = 0.300 \text{ eV}$$

$$E_i - F_p = kT \ln\left(\frac{p}{n_i}\right) = 0.02585 \ln\left(\frac{10^{14}}{10^{10}}\right) = 0.02585 \ln(10^4) = 0.02585 \times 9.210 = 0.238 \text{ eV}$$

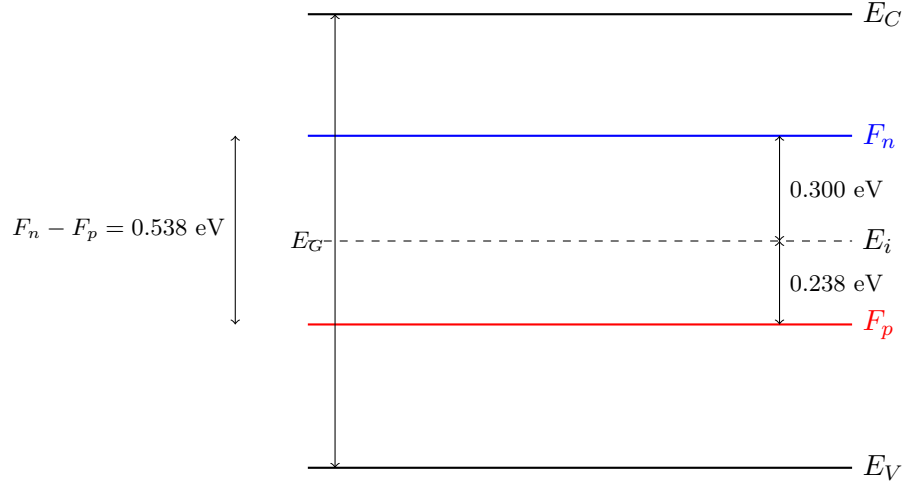


Figure 1: Band diagram under uniform illumination. F_n is above E_i (electrons are majority carriers), and F_p drops below E_i due to the large excess hole concentration.

Problem 4

Given: Si, slope of $F_p = 10^3$ eV/cm, $n_i = 10^{10}$ cm⁻³, $T = 300$ K.

From the band diagram: F_n is **flat** (horizontal), F_p has a positive slope (increasing with x), and E_i is flat at the reference energy level.

Setup

The current densities in terms of quasi-Fermi levels are:

$$J_n = qn\mu_n \frac{dF_n}{dx}, \quad J_p = qp\mu_p \frac{dF_p}{dx}$$

where F_n and F_p are in eV (the numerical value in eV/cm equals V/cm).

Electron current

Since F_n is flat:

$$\frac{dF_n}{dx} = 0 \implies J_n = 0$$

Hole current

From the graph at $x = 0$, F_p appears to be approximately at E_i , so:

$$p(0) = n_i \exp\left(\frac{E_i - F_p}{kT}\right) = n_i \exp(0) = n_i = 10^{10} \text{ cm}^{-3}$$

Using $\mu_p = 480$ cm²/Vs (standard Si value at 300 K):

$$\begin{aligned} J_p &= qp\mu_p \frac{dF_p}{dx} = (1.6 \times 10^{-19})(10^{10})(480)(10^3) \\ &= (1.6 \times 10^{-19})(4.8 \times 10^{15}) \\ J_p &= 7.68 \times 10^{-4} \text{ A/cm}^2 \end{aligned}$$

Total current at $x = 0$

$$J = J_n + J_p = 0 + 7.68 \times 10^{-4}$$

$J(x = 0) \approx 7.68 \times 10^{-4} \text{ A/cm}^2 \quad (\text{in the } +x \text{ direction})$

The current is positive (in the $+x$ direction) because $dF_p/dx > 0$.

Note: If a different value of $F_p(0)$ is read from the graph, substitute the appropriate $p(0) = n_i \exp[(E_i - F_p(0))/(kT)]$ into the formula. Similarly, if μ_p is specified differently, adjust accordingly.

Problem 5

Given: $E_G = 1$ eV, $n_i = 10^{10}$ cm⁻³, $\mu_n = \mu_p = 1000$ cm²/Vs, $T = 300$ K.

From the band diagram:

- Region I ($0 \leq x \leq 1$ μm): Flat bands. At $x = 0$: $E_C - E_F = E_G/4 = 0.25$ eV.
- Region II ($1 \leq x \leq 3$ μm): Bands vary linearly (slope downward).
- Region III ($3 \leq x \leq 4$ μm): Flat bands. At $x = 4$: $E_F - E_i = E_G/4 = 0.25$ eV.

Key energy relations at $x = 0$:

$$E_C(0) - E_F = \frac{E_G}{4} = 0.25 \text{ eV}$$

Since $E_C - E_i = E_G/2 = 0.5$ eV:

$$E_F - E_i(0) = E_C(0) - E_i(0) - [E_C(0) - E_F] = 0.5 - 0.25 = 0.25 \text{ eV}$$

From the diagram, the total drop in E_i (and E_C , E_V) from the left flat region to the right flat region is $\Delta E = 0.5$ eV. This drop occurs linearly over the region $1 \mu\text{m} \leq x \leq 3 \mu\text{m}$.

(a) Electrostatic potential $V(x)$

The electrostatic potential is related to the intrinsic level by:

$$V(x) = -\frac{E_i(x) - E_i(0)}{q}$$

Taking $V(0) = 0$:

- **Region I** ($0 \leq x \leq 1$ μm): Bands flat $\Rightarrow V = 0$.
- **Region II** ($1 \leq x \leq 3$ μm): E_i decreases linearly by 0.5 eV over 2 μm :

$$V(x) = \frac{0.5}{2 \mu\text{m}}(x - 1 \mu\text{m}) = 0.25(x - 1) \text{ V} \quad (x \text{ in } \mu\text{m})$$

- **Region III** ($3 \leq x \leq 4$ μm): Bands flat $\Rightarrow V = 0.5$ V.

$V(x = 4 \mu\text{m}) = 0.5 \text{ V}$

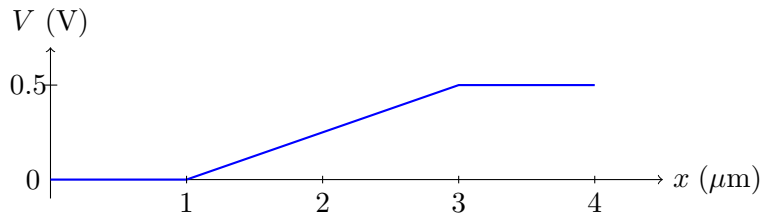


Figure 2: Electrostatic potential $V(x)$ inside the semiconductor.

(b) Electric field $\mathcal{E}(x)$

$$\mathcal{E} = -\frac{dV}{dx}$$

• **Region I** ($0 < x < 1 \mu\text{m}$): $\mathcal{E} = 0$.

• **Region II** ($1 < x < 3 \mu\text{m}$):

$$\frac{dV}{dx} = \frac{0.5 \text{ V}}{2 \times 10^{-4} \text{ cm}} = 2500 \text{ V/cm}$$

$$\mathcal{E} = -2500 \text{ V/cm}$$

(The field points in the $-x$ direction.)

• **Region III** ($3 < x < 4 \mu\text{m}$): $\mathcal{E} = 0$.

$$\boxed{\mathcal{E}(x = 2 \mu\text{m}) = -2500 \text{ V/cm}}$$

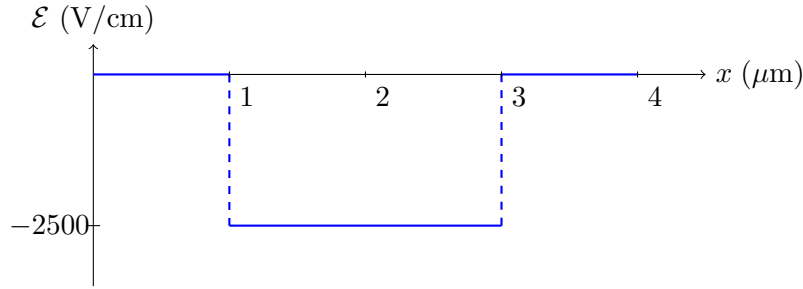


Figure 3: Electric field $\mathcal{E}(x)$ inside the semiconductor.

(c) Carrier concentrations $\log_{10}(n)$ and $\log_{10}(p)$

Using $n = n_i \exp\left(\frac{E_F - E_i}{kT}\right)$ and $p = n_i \exp\left(\frac{E_i - E_F}{kT}\right)$:

At $x = 0$ (Region I):

$$E_F - E_i = 0.25 \text{ eV}$$

$$n(0) = 10^{10} \exp\left(\frac{0.25}{0.02585}\right) = 10^{10} \times e^{9.671} = 10^{10} \times 1.59 \times 10^4 = 1.59 \times 10^{14}$$

$$\log_{10}(n(0)) \approx 14.2$$

$$p(0) = \frac{n_i^2}{n(0)} = \frac{10^{20}}{1.59 \times 10^{14}} = 6.3 \times 10^5$$

$$\log_{10}(p(0)) \approx 5.8$$

At $x = 4 \mu\text{m}$ (Region III):

$E_F - E_i = 0.25 \text{ eV}$ (from diagram), so:

$$n(4) = 1.59 \times 10^{14}, \quad \log_{10}(n) \approx 14.2$$

$$p(4) = 6.3 \times 10^5, \quad \log_{10}(p) \approx 5.8$$

In the linear region (1–3 μm), if E_F and E_i change at the same rate, then $E_F - E_i$ remains constant and n and p are uniform throughout. If E_F is constant (equilibrium), then $E_F - E_i$ increases in the middle (since E_i drops), giving higher n and lower p in the sloped region.

$$\log_{10}(n) \approx 14.2, \quad \log_{10}(p) \approx 5.8 \quad \text{at } x = 0 \text{ and } x = 4 \mu\text{m}$$

(c) Is the semiconductor in equilibrium?

For a system in equilibrium, the Fermi level E_F must be **constant** (independent of position) throughout the semiconductor, and no net current flows.

From the band diagram, a single Fermi level E_F is shown. If E_F is perfectly flat (constant) while the bands tilt, the system is in **equilibrium** — the built-in field from non-uniform doping exactly balances diffusion, and no net current flows.

However, if a potential is applied externally (as suggested by the problem statement), then E_F would vary with position and the system would **not** be in equilibrium.

If equilibrium: $E_F = \text{constant}$, drift and diffusion currents for each carrier type cancel exactly, and $J_n = J_p = 0$.

If not in equilibrium: An applied voltage causes band tilting, E_F varies with position, and current flows through the device.

The semiconductor is in equilibrium if E_F is constant (flat) throughout.

The presence of a single E_F (not split into F_n and F_p) suggests equilibrium. In equilibrium, $J = 0$ everywhere because drift exactly cancels diffusion for each carrier.

(d) Diffusion currents at $x = 2 \mu\text{m}$

Using the Einstein relation: $D_n = D_p = \frac{kT}{q}\mu = (0.02585)(1000) = 25.85 \text{ cm}^2/\text{s}$.

If the system is in equilibrium (E_F constant), then as E_i drops, $E_F - E_i$ increases linearly in Region II. At $x = 2 \mu\text{m}$ (midpoint):

$$E_F - E_i(x = 2) = 0.25 + \frac{0.5}{2} \times \frac{(2 - 1)}{2} = 0.25 + 0.125 = 0.375 \text{ eV}$$

n and p vary with position in Region II, so $dn/dx \neq 0$ and $dp/dx \neq 0$.

Since $n = n_i e^{(E_F - E_i)/(kT)}$ and E_F is constant while E_i varies:

$$\frac{dn}{dx} = n \cdot \frac{1}{kT} \left(-\frac{dE_i}{dx} \right)$$

With $dE_i/dx = -0.5/(2 \times 10^{-4}) = -2500 \text{ eV/cm}$:

$$\frac{dn}{dx} = n \cdot \frac{2500}{0.02585} = n \times 9.671 \times 10^4 \text{ cm}^{-1}$$

At $x = 2 \mu\text{m}$: $n = 10^{10} e^{0.375/0.02585} = 10^{10} \times e^{14.51} = 10^{10} \times 1.99 \times 10^6 = 1.99 \times 10^{16} \text{ cm}^{-3}$

Electron diffusion current:

$$\begin{aligned} J_{n,\text{diff}} &= qD_n \frac{dn}{dx} = (1.6 \times 10^{-19})(25.85)(1.99 \times 10^{16})(9.671 \times 10^4) \\ &= (1.6 \times 10^{-19})(25.85)(1.924 \times 10^{21}) \end{aligned}$$

$$J_{n,\text{diff}} = 7.96 \text{ A/cm}^2 \quad (\text{in the } +x \text{ direction})$$

Hole concentration at $x = 2 \mu\text{m}$:

$$p = \frac{n_i^2}{n} = \frac{10^{20}}{1.99 \times 10^{16}} = 5.03 \times 10^3 \text{ cm}^{-3}$$

$$\frac{dp}{dx} = p \cdot \frac{1}{kT} \frac{dE_i}{dx} = p \cdot \frac{-2500}{0.02585} = -p \times 9.671 \times 10^4 \text{ cm}^{-1}$$

Hole diffusion current:

$$J_{p,\text{diff}} = -qD_p \frac{dp}{dx} = -qD_p \times p \times (-9.671 \times 10^4) = qD_p \cdot p \cdot 9.671 \times 10^4$$

$$= (1.6 \times 10^{-19})(25.85)(5.03 \times 10^3)(9.671 \times 10^4)$$

$$\boxed{J_{p,\text{diff}} = 2.01 \times 10^{-9} \text{ A/cm}^2 \quad (\text{in the } +x \text{ direction})}$$

(e) Drift currents at $x = 2 \mu\text{m}$

$\mathcal{E} = -2500 \text{ V/cm}$ at $x = 2 \mu\text{m}$.

Electron drift current:

$$J_{n,\text{drift}} = qn\mu_n\mathcal{E} = (1.6 \times 10^{-19})(1.99 \times 10^{16})(1000)(-2500)$$

$$\boxed{J_{n,\text{drift}} = -7.96 \text{ A/cm}^2 \quad (\text{in the } -x \text{ direction})}$$

Hole drift current:

$$J_{p,\text{drift}} = qp\mu_p\mathcal{E} = (1.6 \times 10^{-19})(5.03 \times 10^3)(1000)(-2500)$$

$$\boxed{J_{p,\text{drift}} = -2.01 \times 10^{-9} \text{ A/cm}^2 \quad (\text{in the } -x \text{ direction})}$$

Note: As expected for equilibrium, $J_{n,\text{drift}} + J_{n,\text{diff}} = 0$ and $J_{p,\text{drift}} + J_{p,\text{diff}} = 0$. The drift and diffusion components exactly cancel for each carrier type.

(f) Minimum kinetic energy for electron to move from $x = 4$ to $x = 0$

An electron at $x = 4$ sits at the conduction band edge $E_C(4)$. To move to $x = 0$ without changing its total energy, it must have enough kinetic energy to overcome the potential energy barrier.

Since the bands are higher at $x = 0$ (the bands slope downward from left to right):

$$E_C(0) - E_C(4) = E_i(0) - E_i(4) = 0.5 \text{ eV}$$

The electron must have a minimum kinetic energy equal to the increase in potential energy:

$$\boxed{KE_{\min} = E_C(0) - E_C(4) = 0.5 \text{ eV}}$$